



# Rounding integers to one significant figure

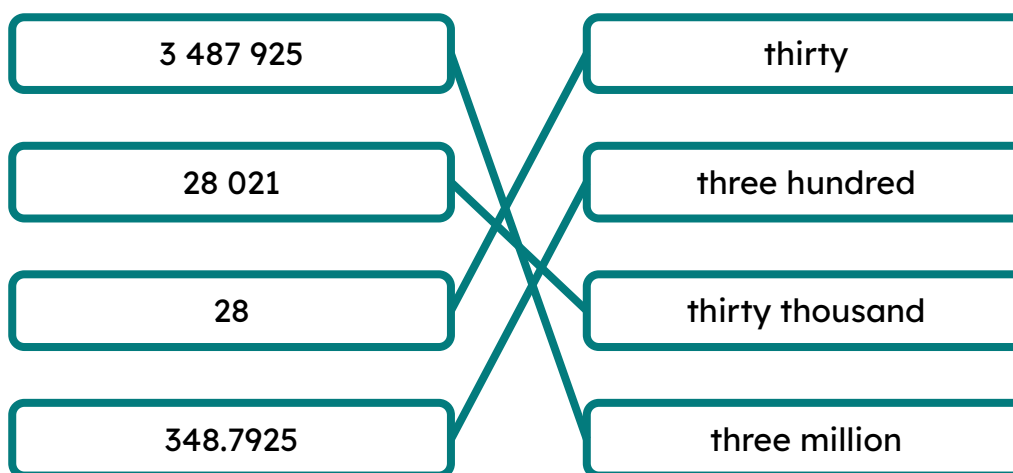
## Task A: Identifying significant figures

Circle the first **significant figure** in each of the following:

- a) 150 267
- b) 35604 987 932
- c) 60 253 461
- d) 137
- e) 98 200
- f) 4 987 932

## Task B: Using a number line for significant figures

1) Match the following to the correct rounded to one significant figure number:



2) Complete the following:

| Number    | The 1st significant figure is in the _____ column. | Round to the nearest _____. | Number line                              | To 1 s.f. |
|-----------|----------------------------------------------------|-----------------------------|------------------------------------------|-----------|
| 36        | tens                                               | 10                          | 30      35      40<br>↓                  | 40        |
| 832       | hundreds                                           | 100                         | 800      850      900<br>↓               | 800       |
| 3519      | thousands                                          | 1000                        | 3000      3500      4000<br>↓            | 4000      |
| 43 867.09 | 10 thousands                                       | 10 000                      | 40 000      45 000      50 000<br>↓      | 40 000    |
| 992 782   | 100 thousands                                      | 100 000                     | 900 000      950 000      1 000 000<br>↓ | 1 000 000 |

## Task C: Rounding to one significant figure

1) Write each of the numbers below to one significant figure, find the answer in the grid to complete this joke:

A sheepdog goes out to get the farmer's sheep in. He comes back and says to the farmer, 'that's all 100 sheep in the barn' The farmer says, 'that's odd we only had 97 this morning'. The sheepdog replies ...

| A   | B    | C    | D      | E    | F      | G      | H   | I    | J    | K      | L    | M      |
|-----|------|------|--------|------|--------|--------|-----|------|------|--------|------|--------|
| 200 | 60   | 400  | 30 000 | 10   | 20 000 | 700    | 90  | 5000 | 20   | 100    | 9000 | 80 000 |
| N   | O    | P    | Q      | R    | S      | T      | U   | V    | W    | X      | Y    | Z      |
| 70  | 2000 | 7000 | 40     | 8000 | 300    | 10 000 | 800 | 900  | 1000 | 50 000 | 40   | 600    |

1)  $4682 \approx 5000$  I

2)  $1800 \approx 2000$  O

3)  $1499 \approx 1000$  W

4)  $71 \approx 70$  N

5)  $104 \approx 100$  K

6)  $5498 \approx 5000$  I

7)  $65 \approx 70$

N

8)  $13 \approx 10$

E

9)  $840 \approx 800$

U

10)  $8287 \approx 8000$

R

11)  $34\,999 \approx 30\,000$

D

12)  $1560 \approx 2000$

O

13)  $25\,873 \approx 30\,000$

D

14)  $75\,001 \approx 80\,000$  M

15)  $94 \approx 90$  H

16)  $9810 \approx 10\,000$  T

17)  $13 \approx 10$  E

18)  $7459 \approx 7000$  P

19)  $762 \approx 800$  U

*'I know, I rounded them up!'*

2) What values can the missing digit take?

a)  $4?90 = 4000$  (1 s.f.)

*0, 1, 2, 3 or 4*

b)  $9?12 = 10\,000$  (1 s.f.)

*5, 6, 7, 8, or 9*

c)  $34\,?90 = 30\,000$  (1 s.f.)

*any digit*

d)  $678? = 7000$  (1 s.f.)

*any digit*

